

Unsupervised Exploratory Graph Analysis of Communication and Recommendation Datasets

Project Report

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Abstract

This study presents an exploratory analysis of two large-scale graph datasets from distinct domains: the `email-Eu-core` organizational communication network and the `Amazon0302` product co-purchasing network. The objective is to investigate how structural and behavioral characteristics differ between human communication and machine-driven recommendation systems. Five key graph properties were examined, including *in/out-degree distributions*, *reciprocity*, *node centrality measures*, and the combination of *number of triangles* and *clustering coefficient*. These properties were selected to capture both local and global aspects of connectivity, influence, and cohesiveness. Quantitative results and qualitative interpretations reveal that the `email-Eu-core` network exhibits a dense, reciprocal, and community-oriented structure driven by departmental cohesion, while the `Amazon0302` network is sparse and dominated by directional relationships toward popular hub products. The source code and replication package of this study are publicly available [here](#).

Keywords: Graph analysis, Neo4j, network structure, unsupervised exploratory data analysis.

1 Introduction

Networks are a powerful abstraction for representing relationships among entities across diverse domains. With the growing availability of large-scale graph datasets and mature graph database systems, such as Neo4j, it has become increasingly feasible to perform detailed exploratory studies that reveal both local and global network behaviors and characteristics.

The goal of this study is to conduct an unsupervised exploratory analysis on two large-scale graph datasets drawn from distinct domains: a social communication network and a product co-purchasing network. Specifically, we examine the `email-Eu-core` dataset, representing internal email exchanges within a European research institution, and the `Amazon0302` dataset, representing product co-purchases on the Amazon website. Together, these datasets enable us to investigate how structural and behavioral properties differ between human communication and consumer recommendation systems.

Our analysis focuses on five fundamental graph properties that collectively characterize network structure and interaction dynamics: (i) in/out-degree distributions, (ii) reciprocity, (iii) node centrality measures, and (iv) the combination of number of triangles and clustering coefficient.

Through our analysis, we found that the `email-Eu-core` network exhibits a moderately dense but uneven structure, driven by a few highly active individuals. Communication within departments is strongly reciprocal and cohesive, while cross-department interaction remains limited, forming a collaborative yet compartmentalized organizational pattern. In contrast, the `Amazon0302` co-purchasing network is sparse and asymmetric, dominated by a small number of high in-degree “hub” products that attract numerous incoming links. These products serve as central sinks in the network, while most others connect to only a few related items.

2 Methodology and Implementation

In this section of the report, we discuss the implementation details and method we adapted in our work.

2.1 Dataset Selection and Characteristics

For our comparative exploratory graph analysis, we selected two publicly available large-scale graph datasets from distinct domains to satisfy the project requirement of cross-domain comparison. Here is a summary of each dataset, including their domain context, structural characteristics, and key statistics.

2.1.1 Email-Eu-core Dataset

The `email-Eu-core` dataset comes from the communication network within a large European research institution, provided by the Stanford Network Analysis Project (SNAP). The graph has person nodes and each directed edge (u, v) between these nodes indicates that person u sent an email to person v within the internal network of the institution. Upon inspection of the dataset, we found that no duplicate directed edges exist between any two nodes, meaning there are no multiple emails recorded from one sender to the same receiver. In addition to the communication links, the dataset includes community labels, where each person belongs to exactly one of 42 departments within the institution. To incorporate this structural information into our analysis, we extracted the department labels and modeled each department as a separate node, connecting every person node to their corresponding department node. Our intuition was that, because the graph originates from an internal organizational email network, it inherently captures human communication patterns, departmental structures, and organizational hierarchy, while also reflecting underlying social groups and workflow connectivity within the institution. Table 1a shows the key stats for this dataset. Before importing the data into Neo4j, we performed light preprocessing to convert the original text files into CSV format suitable for graph modeling. The first file extracted person–department mappings, resulting in a table with columns $(person_id, department_id)$, while the second captured email communications, formatted as (src, dst) to represent sender–receiver relationships. These transformations produced two clean CSV files ready for import.

2.1.2 Amazon Product Co-Purchasing Network

This dataset represents a product co-purchasing network drawn from the Amazon Website. If a product i is frequently co-purchased with product j , then there is a directed edge from i to j . Figure 1 illustrates the structure of the Amazon product co-purchasing network. Each node represents a product, and a directed edge from product i to product j indicates that i is frequently purchased together with j . The direction of the arrow captures the co-purchasing tendency, while the density and connectivity of the network reveal

clusters of products that are often bought in combination. Such a representation provides insight into consumer behavior patterns and can be leveraged for recommendation systems.

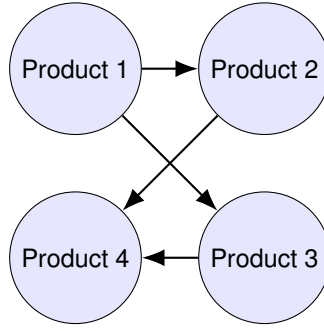


Figure 1: Example device connection network.

In this network nodes are products and edges represent purchase-association relationships. Reflects how products are jointly considered by buyers (i.e., a network of consumer behavior), rather than human social connections. Table 1b shows the key stats for this dataset. The graph edges for this dataset are publicly available in text format. To construct the graph, we first extracted the unique product IDs appearing in the edge list and used them to create the corresponding product nodes. We then used the edges CSV file to establish the directed recommendation relationships between them. This process yielded a clean, directed product co-purchase graph suitable for further analysis.

Graph Property	Value
Person Nodes	1,005
Edges (Emails sent)	25,571
Department Nodes	42

(a) email-Eu-core dataset

Graph Property	Value
Nodes (Products)	262,111
Edges (Co-purchases)	1,234,877

(b) Amazon0302 dataset

Table 1: Key statistics for the email-Eu-core and Amazon0302 graph datasets.

2.2 Graph Properties selection

In this section, we present the five graph properties examined in our comparative analysis: in/out-degree distributions, reciprocity, node centrality measures, and the combination of number of triangles and clustering coefficient. Each property offers a distinct structural perspective on the networks, from local connectivity patterns to global hierarchical organization.

2.2.1 In/Out-Degree Distributions

The degree of a node quantifies how many connections it has. In directed graphs, this splits into in-degree (number of incoming edges) and out-degree (number of outgoing edges). The degree distribution captures how these values are distributed across the network and is fundamental in identifying structural patterns such as hubs and hierarchies. We selected this property because it provides a first-order view of network connectivity, helping reveal whether the network is uniform (most nodes having similar degree) or skewed (few highly connected nodes dominating the structure).

To compute degree distributions, we used the regular Neo4j Cypher queries, extracting in-degree and out-degree separately for each node and aggregating their statistical summaries (mean, median, standard deviation, min, and max). The full queries are available in our replication package.

2.2.2 Reciprocity

Reciprocity measures the proportion of edges that are bidirectional, i.e., when node A links to B and B links back to A. It captures the degree of mutual interaction in a directed network. We included this property because it will showcase the behavior of top senders and recipients in the email dataset. This graph property will also highlight reciprocal recommendations in the amazon dataset, revealing more about its overall structure. We calculated global and local reciprocity using a custom Cypher query that counts mutual edge pairs across all nodes for both of our datasets.

2.2.3 Node Centrality Measures

Centrality quantifies the importance or influence of nodes within a network. We focus on degree centrality (how connected a node is) and PageRank (influence based on the importance of connected neighbors). This property highlights leadership or influence roles in communication and key connector products in recommendation systems.

We computed degree centrality directly from in/out-degree values similar to our queries for the pure in/out-degree graph property analysis. We also calculated PageRank for our nodes using the Neo4j Graph Data Science library, particularly, using the `gds.pageRank.stream()` algorithm with a damping factor of 0.85 and 20 max iterations.

2.2.4 Number of Triangles and Clustering Coefficient

A triangle represents a set of three nodes where each node is connected to the other two, and the clustering coefficient measures the likelihood that two neighbors of a node are connected, quantifying the tendency of a network to form tightly knit groups. We analyzed these two properties jointly because they are complementary indicators of cohesiveness. To compute them, we first converted the original graph into an undirected version, since both metrics rely on mutual connectivity between nodes. We then used the `gds.triangleCount.stream()` and the approximated `gds.localClusteringCoefficient.stream()` algorithms in our custom Cypher queries to obtain triangle counts and clustering coefficients, respectively.

2.3 Experimental Environment Setup

All experiments were conducted using Neo4j Desktop (version 2.0.4) running on an Ubuntu machine. For data management and isolation, we created two separate databases, one for each dataset.

Within Neo4j Desktop, we utilized the built-in data import feature to load our preprocessed CSV files into graph form. Before importing, we defined the schema for each database by explicitly specifying the node types and relationship types that characterize the data model. For the email dataset, this included Person and Department nodes connected via SENT_EMAIL and IS_IN relationships. For the Amazon dataset, we defined a single node type, Product, connected by the ALSO_BOUGHT relationship.

Once the schema definitions were complete, the corresponding CSV files were imported into Neo4j, automatically generating the nodes and relationships according to the defined schema.

3 Experimental Results and Analysis

In this section, we present the experimental analysis conducted on both datasets based on the graph properties introduced earlier. We analyze each property quantitatively and qualitatively to extract meaningful structural insights from the networks.

3.1 In/Out-Degree Distributions

3.1.1 Email Dataset Results

The degree distributions for the Email dataset are shown in Table 2a and visualized in Figure 2a. Overall, the email dataset forms a moderately dense structure, with each person connected to roughly 25 others on average. The average in-degree and out-degree are nearly identical (25.44), which is expected since each email creates one outgoing and one incoming connection. The degree distribution, however, is highly uneven and right-skewed. While the median person sends emails to about 15 others and receives messages from around 18, the standard deviation of degrees (28 for out-degree and 33 for in-degree) is even higher than the average itself, confirming a wide spread in communication activity. A few highly active individuals dominate the network, with some sending messages to over 300 recipients, and others receiving from more than 200 unique senders. This imbalance reveals a hierarchical structure, where a small number of people (likely managers, coordinators, or administrative staff) broadcast information to many, while most individuals communicate within smaller circles.

At the department level, the average size is about 24 people, but the standard deviation (24) shows that sizes vary widely, some departments have only one or two members, while others exceed a hundred. Departments also differ in how internally connected they are: the average number of internal emails is about 227, ranging from 1 to over 1,500. In contrast, cross-department communication is much lower, with departments exchanging only about 13 emails on average with others. This pattern reflects strong internal cohesion and limited inter-department collaboration.

Finally, the node activity distribution shows 137 people who never sent an email (zero out-degree) and 14 who never received one (zero in-degree). Combined with the reported standard deviation and median sent and received emails, this supports the trend that people generally receive more distinct messages than they send.

Summary: The email network is moderately dense but highly uneven, with a few “super-senders” and “super-receivers” dominating communication. Large standard deviations in degree values confirm a hierarchical structure where most people interact within small circles and generally receive more distinct messages than they send. Departments show strong internal cohesion and limited cross-department communication, reflecting a compartmentalized organizational pattern.

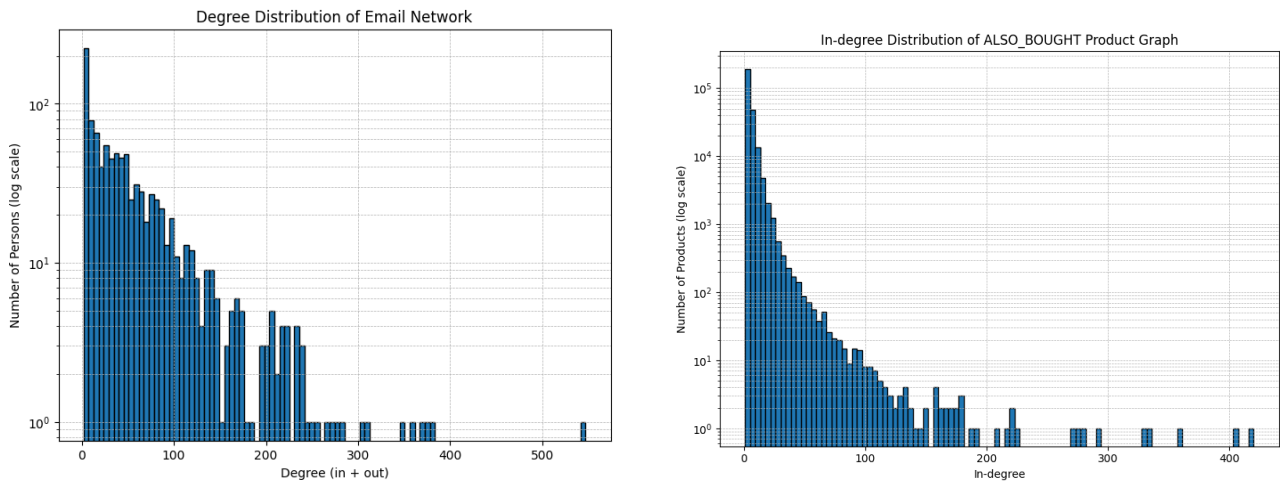
3.1.2 Amazon Dataset Results

The degree distributions for the Amazon dataset are shown in Table 2b and visualized in Figure 2b. In the Amazon dataset, the average in-degree (4.71) and average out-degree (4.71) are numerically similar, but the underlying distributions differ sharply. The maximum in-degree (420) shows that some products are recommended by hundreds of others, while the maximum out-degree (5) indicates that each product recommends only a handful of others. The standard deviation values reinforce this imbalance: the in-degree variation (5.71) is much higher than the out-degree variation (0.95), revealing a strong asymmetry where a few products act as popular hubs that attract many incoming links, whereas most products point outward to only a few related items. This pattern produces a highly right-skewed distribution, meaning that while most products have few connections, a small subset accumulates a large share of incoming recommendations. The median degrees (in = 3, out = 5) are lower than the averages, confirming that the majority of products are relatively isolated or niche, while a minority dominate the co-purchase network. The imbalance between in- and out-degree behavior reflects the nature of recommendation systems: edges tend to flow toward popular or complementary products. As a result, common items like chargers, cables, or accessories serve as central nodes recommended by many others, while most products act as peripheral sources.

Summary: The Amazon co-purchase graph is sparse and dominated by a few high in-degree “hub” products that are frequently recommended by others. Low out-degree variance and higher in-degree variance confirm a popularity-driven structure where most products link to a few common items. This dataset exhibits a clear sink-oriented hierarchy, typical of consumer recommendation systems.

3.1.3 Comparing the Datasets

The email network exhibits a moderately dense and highly reciprocal structure, where a few “super-senders” and “super-receivers” dominate communication, while most individuals exchange messages within small, localized circles. The close balance between in- and out-degrees reflects the two-way nature of human communication. In contrast, the Amazon co-purchase graph is sparse and characterized by a one-directional flow toward a few popular “hub” products that attract many incoming links.



(a) Degree distribution of the email communication network (in + out). The plot reveals a highly right-skewed distribution, where most individuals exchange emails with a small number of others, while a few “super-senders” and “super-receivers” connect to hundreds of unique peers. This imbalance reflects a hierarchical communication structure within the organization.

(b) In-degree distribution of the Amazon ALSO_BOUGHT product graph. The distribution is heavily right-skewed, showing that a small subset of products serve as popular hubs recommended by many items. This structure highlights the sink-oriented nature of product recommendation networks.

Figure 2: Degree distributions of the two analyzed datasets.

Statistic	Value
Department Stats	
Avg. people per department	23.93
Min / Max size	1 / 109
Std. dev. (dept. size)	24.22
Internal Emails	
Avg. internal emails	227.50
Min / Max	1 / 1,506
External Emails	
Avg. inter-dept. emails	13.54
Min / Max	1 / 229
Degree Stats	
Avg. in/out degree	25.44
Max in / out	212 / 334
Min in / out	0 / 0
Median in / out	18 / 15
Std. dev. (in / out)	33.32 / 27.97
Zero out-degree nodes	137
Zero in-degree nodes	14

(a) Email dataset

Statistic	Value
Degree Stats	
Avg. in/out degree	4.71
Max in / out	420 / 5
Min in / out	1 / 0
Median in / out	3 / 5
Std. dev. (in / out)	5.71 / 0.95

(b) Amazon dataset

Table 2: Comparison of degree-related statistics for the email-Eu-core and Amazon0302 datasets.

3.2 Reciprocity

3.2.1 Email Dataset Results

We computed this property only for Person nodes and the edges connecting them, excluding Department nodes. The reciprocity stats for this dataset are shown in Table 3a. The overall reciprocity of the email network is 0.72, meaning roughly 72% of all email interactions are mutual. I.e., if one person emails another, a reply is likely. This high value aligns with the moderately dense structure observed earlier, confirming that frequent and bidirectional exchanges form the backbone of this communication network.

When focusing on the top 20 senders, the average reciprocity slightly increases to 0.77, while the top 20 receivers show an exceptional 0.99. These values reinforce the earlier finding of “super-senders” and “super-receivers” who dominate the network. Despite their central roles, these high-degree individuals are

not broadcasting in isolation; they maintain active, two-way correspondence with many others.

Between departments, reciprocity falls to 0.29, confirming a strong internal vs. external divide also seen in the degree analysis. The low inter-department reciprocity suggests that when cross-team communication occurs, it is mostly one-directional, consistent with reporting chains or formal announcements.

Summary: The email network demonstrates high overall reciprocity within and across individuals, confirming a conversational and collaborative environment. Yet, this reciprocity decreases significantly across departmental boundaries, underscoring an internally cohesive but organizationally segmented communication structure.

3.2.2 Amazon Dataset Results

The reciprocity stats for this dataset are shown in Table 3b. The global reciprocity of the Amazon co-purchase network is approximately 0.54, indicating that just over half of all “also-bought” relationships are mutual. The relatively high standard deviation (0.29) reveals large variability across products: some participate in strongly reciprocal co-buy patterns, while others exist in one-directional recommendation flows.

When focusing on the top 20 products by in-degree, those receiving the most recommendations, we observe an average reciprocity of 0.75. This suggests that highly popular products tend to be more reciprocal. This increase in reciprocity at the bottom of the sink and the high standard deviation for global reciprocity suggests that products at the top of the sink don’t reciprocate and their outgoing edges seem to be downward.

Summary: The Amazon dataset displays selective reciprocity rather than uniform bidirectionality. Mutual “also-bought” relationships are concentrated among popular and complementary products. Meanwhile, the majority of products participate in one-directional recommendation flows, linking outward to well-known or high-demand items.

3.2.3 Comparing the Datasets

The email network shows high and balanced reciprocity, reflecting conversational, two-way communication within departments but weaker cross-department exchange. In contrast, the Amazon network exhibits moderate and uneven reciprocity, concentrated among popular products, while a substantial number of links remain one-directional.

Statistic	Value
Global reciprocity	0.72
Avg. reciprocity (top 20 senders)	0.77
Avg. reciprocity (top 20 receivers)	0.99
Avg. cross-department reciprocity	0.29

(a) email-Eu-core dataset

Statistic	Value
Global reciprocity	0.54
Std. dev. (global reciprocity)	0.29
Avg. reciprocity (top 20 in-degree)	0.75

(b) Amazon0302 dataset

Table 3: Comparison of reciprocity statistics for the email-Eu-core and Amazon0302 datasets.

3.3 Node Centrality Measures

3.3.1 Email Dataset Results

Department-level degree centrality: In 27 of 42 departments, the same individual is both the top sender (out-degree) and the top receiver (in-degree), effectively making them the department’s degree-central hub. For the other 15 departments, we selected the top total-degree person (the one with the most edges overall, in + out) and found that their send–receive imbalance is small. This means that even where the single most active sender and the single most active receiver are different people, the department’s most connected person still participates in communication nearly symmetrically. This further strengthens the claims made in the reciprocity analysis section. In practice, this aligns with roles like team leads or coordinators who broadcast updates but also field many replies, keeping their net traffic balanced.

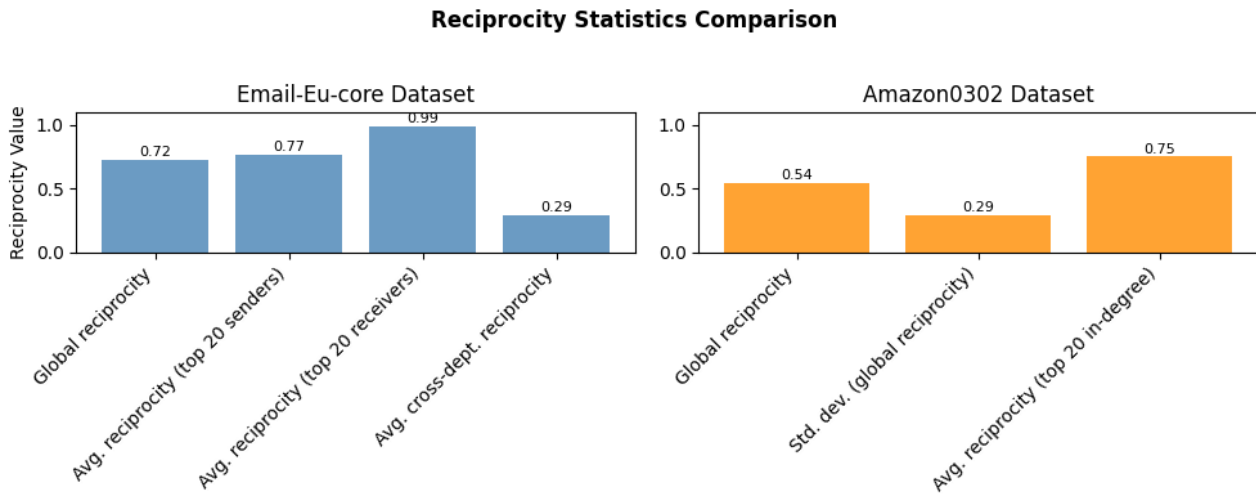


Figure 3: Reciprocity of the network

PageRank vs. degree: To capture influence beyond raw activity, we computed PageRank and compared its top 20 with the top 20 by total degree. The overlap is 14/20 (70%), indicating that most high-volume communicators are also structurally influential (connected to other important communicators). This pattern reflects a hierarchical communication structure within departments, where the most influential people interact primarily with one another, forming tightly knit cores of information exchange. Such overlap between activity and influence suggests that departmental communication is organized around well-defined roles and leadership structures rather than random interactions.

Summary: In most departments, a single central figure dominates both sending and receiving activity, and even where leadership roles differ, the overall exchange remains nearly symmetrical. The strong overlap between PageRank and degree centrality reinforces this pattern, showing that the most active communicators tend to interact primarily with one another. Together, these results reveal that departmental communication is shaped by clear organizational hierarchies.

3.3.2 Amazon Dataset Result

the Amazon co-purchase network shows a pronounced separation between activity and influence. Only 30% of the top ten products by degree centrality also appear among the top PageRank nodes. This low overlap indicates that sink products, such as accessories or universally compatible items, receive links from many others but rarely point to one another. As a result, they accumulate numerous direct recommendations but contribute little to the overall flow of influence within the network. PageRank, in turn, elevates products that lie between popular clusters, items co-bought across several distinct high-demand categories, making them more influential despite having fewer total links.

Summary: The Amazon network's structure separates popularity from influence. High-degree products dominate through volume, while PageRank highlights cross-category connectors that shape the network's global flow.

3.3.3 Comparing the Datasets

In the email network, activity and influence largely coincide, most central communicators are both highly active and structurally influential. In contrast, the Amazon network separates popularity from influence: high-degree hub products dominate through volume, while PageRank highlights cross-category connectors that drive global influence across the marketplace.

3.4 Number of Triangles and Clustering Coefficient

3.4.1 Email Dataset Results

The triangle and CC stats for this dataset can be seen in Table 4a. We computed these properties only for Person nodes and the edges connecting them, excluding Department nodes. The email network

contains 105,461 triangles, with an average clustering coefficient of 0.39 and a standard deviation of 0.23. This indicates that group-based communication is common but unevenly distributed across the organization. Given that inter-department communication is relatively low, most triangles are likely formed within departments rather than between them.

Summary: the triangle structure reflects a collaborative but compartmentalized communication network, which is strongly cohesive within teams but weakly connected across them.

3.4.2 Amazon Dataset Results

The triangle and CC stats for this dataset can be seen in Table 4b. The Amazon co-purchase graph contains 717,719 triangles, indicating a moderate level of local transitivity within its recommendation structure. The average clustering coefficient (0.42) shows that about 42% of all possible product triplets form closed “also-bought” groups, while the high standard deviation (0.27) reveals that clustering is highly uneven: some product neighborhoods are tightly interconnected, while others remain sparse. This imbalance becomes clearer when examining high-degree nodes. The top-20 in-degree products have an extremely low average clustering of 0.008, showing that these popular hub items receive links from many unrelated products that rarely connect to one another. This further supports the sink structure proposed in prior sections.

Summary: Clustering in the Amazon co-purchase network reflects a hierarchical sink structure. While most products form locally dense clusters of mutually co-bought items, high in-degree hub products sit at the bottom of the hierarchy, receiving links from many unrelated items that rarely connect to each other.

3.4.3 Comparing the Datasets

The network reflects a collaborative yet compartmentalized structure: strongly cohesive within teams but weakly connected across departments. In contrast, the Amazon network’s clustering reflects a sink structure, where locally dense product groups coexist with sink products that exhibit almost no clustering, connecting otherwise unrelated items.

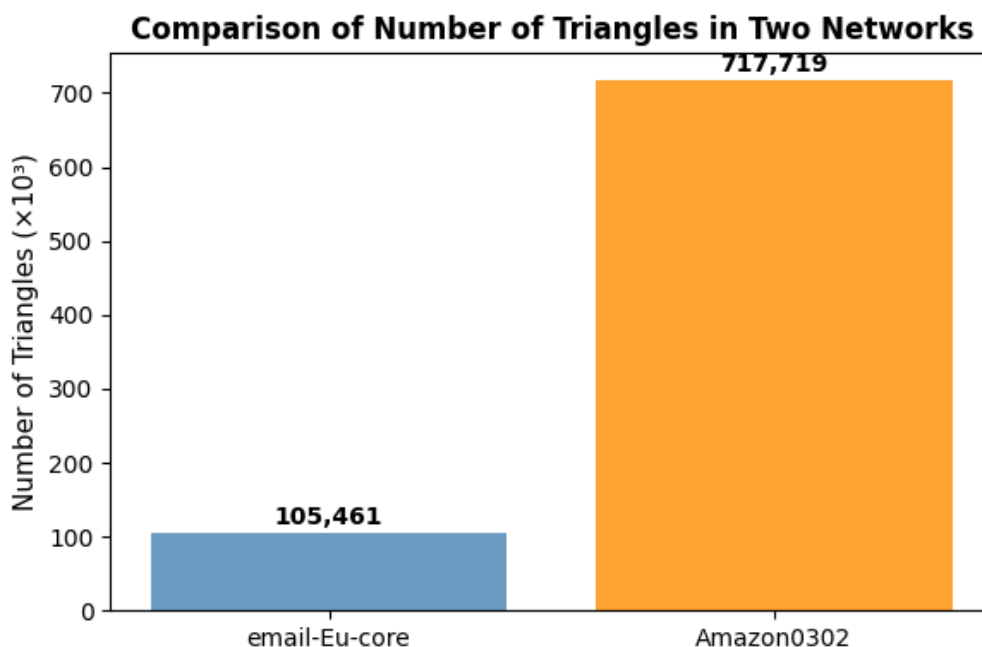


Figure 4: Number of triangles

Statistic	Value
Number of triangles	105,461
Avg. clustering coefficient	0.39
Std. clustering coefficient	0.23
Avg. dept. clustering coefficient	0.36
Std. dev. (dept. clustering)	0.15

(a) email-Eu-core dataset

Statistic	Value
Number of triangles	717,719
Avg. global clustering	0.42
Std. global clustering	0.27
Avg. clustering (top 20 in-degree)	0.008

(b) Amazon0302 dataset

Table 4: Comparison of triangle and clustering coefficient statistics for the email-Eu-core and Amazon0302 datasets.

4 Conclusions

In this study, we conducted a comprehensive exploratory analysis of two large-scale graph datasets drawn from distinct domains: the email-Eu-core organizational communication network and the Amazon0302 product co-purchasing network. The objective was to examine how structural properties differ between a social graph and a recommendation graph. Our investigation focused on five key graph properties that collectively capture both local and global network behaviors: (1) in/out-degree distributions, (2) reciprocity, (3) node centrality measures, and (4) the combination of number of triangles and clustering coefficient. These properties enabled us to uncover both quantitative patterns and qualitative insights about the underlying networks.

Through this analysis, we observed that the email-Eu-core network exhibits a moderately dense yet uneven structure, characterized by a small number of highly active individuals (“super-senders” and “super-receivers”) surrounded by many low-degree participants. High reciprocity values confirmed that communication within the institution is highly bidirectional, though this mutual exchange is largely confined within departments, reflecting strong internal cohesion but limited cross-department interaction. The clustering and triangle analysis further supported this finding, revealing that most closed triads and high clustering coefficients occur within departments, indicating a collaborative but compartmentalized communication structure.

In contrast, the Amazon0302 network presented a sparse and asymmetrical topology. Degree and reciprocity analyses revealed a sink-oriented pattern, where a small number of popular products attract the majority of incoming links, while most products point to only a few related items. Centrality results showed a clear separation between popularity and influence, where high-degree “hub” products dominate by volume, whereas PageRank identifies a smaller subset of cross-category connectors that bridge distinct product clusters. The clustering results reinforced this interpretation, showing that while local product communities are tightly knit, hub products exhibit minimal clustering, acting as connectors among otherwise unrelated items.

Overall, this study demonstrates how combining multiple graph-theoretic measures within Neo4j enables a deep understanding of complex networks across domains. By systematically examining connectivity, reciprocity, influence, and cohesiveness, we revealed that human communication networks tend to be dense, reciprocal, and community-oriented, whereas large-scale product networks are sparse and directional.

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A Complete Dataset and Code Implementations

source code to [Project](#)

B Peer Assessment Report

Group Member Contributions

Member	Contribution (%)	Named Contribution
Alireza Hoseinpour	20	Dataset and graph property selection, Query generation, Result Analysis, Writing
Emmanuel Agbeli	20	Dataset and graph property selection, Query generation, Result Analysis, Writing
Rajni Hiroshima	20	Dataset and graph property selection, Query generation, Result Analysis, Writing
Rahubadde De Silva	20	Dataset and graph property selection, Query generation, Result Analysis, Writing
Cadence Litteral	20	Dataset and graph property selection, Query generation, Result Analysis, Writing